FINDINGS

From the datasets derived from the “train_ctrUa4K.csv" file is used to build a decision tree model to determine eligibility process for giving out loan to Applicant.

There are two models in the Python Worksheet which are Decision Tree models built to determine the eligibility process for giving out loan to applicant. The Two models are the Baseline models called DecisionTree while the other model is called Decisiontree2 which is my personal Model. It is to note that both model were build from the normalize datasets in the Python worksheet.

Also, both model were build using the same random state which is 94 (the last two number of my Student ID) and the sample size but different features or columns or details of the sample.

The features from the original normalize dataset are ['Loan_ID', 'Loan_Status', 'Gender', 'Married', 'Dependents' 'Education', 'Self_Employed', 'ApplicantIncome_log', 'CoapplicantIncome_' , 'Loan_Amount_Term', , 'Credit_History', 'Property_Area', 'LoanAmount_log]. There are thirteen variables/ features in the original normalize dataset.

The Baseline model which is the “DecisionTree” was Build using all the above features except these two features which is the 'Loan_ID' and the 'Loan_Status'.

My personal model named DecisionTree2 was built using only 6 of the original features in order to improve the model and it indeed improve the model with a an improved results. The features are 'ApplicantIncome_log', 'CoapplicantIncome_' , 'Loan_Amount_Term', , 'Credit_History', 'Property_Area', 'LoanAmount_log.

From the results, the baseline line DecisionTree model predicted correctly the True “Yes” and the True “No” as 99 and 27 respectively While my Personal ModelDecisionTree2 model predicted correctly the True Yes and the True No as 103 and 30 respectively which is a better identification from the two

From the model DecisionTree2 the evaluation matric predicted the TRUE “YES” and FALSE “YES” as 103 and 27 while the model predicted the TRUE “NO” and FALSE “NO” as 30 and 22 respectively .

This implies that the model DecisionTree2 using the evaluation matrix

- ✓ the model identify 103 applicant who are actually eligible for the loan as are truly eligible for the loan
- ✓ the model identify 30 applicant who are actually not eligible for the loan as are truly not eligible for the loan On the other hand there were few errors
- ✓ the model identify 22 applicant who are actually eligible for the loan meanwhile they are not truly eligible for the loan
- ✓ The model identify 27 applicant who are actually not eligible for the loan as are eligible for the loan .

In all, the model accuracy is high which is 73percent and we can see the False Positive and False Negatives are lower than the true positive and true negatives. And we have more applicants who are eligible for the Loan.

REFERENCE.

Dream Housing Finance company. (2016). train_ctrUa4K Data set. Analytics Vidhya. Available at https://datahack.analyticsvidhya.com/contest/practice-problem-loan prediction-iii/ProblemStatement (Accessed: 1 February 2023).